Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a longitudinal-wave model to distinguish pitch and sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Light moves from a transparent medium with n=1.6 into air (n=1). Find critical angle using $\sin\theta_c = 1/n$. Describe incidence above it in the ideal model.

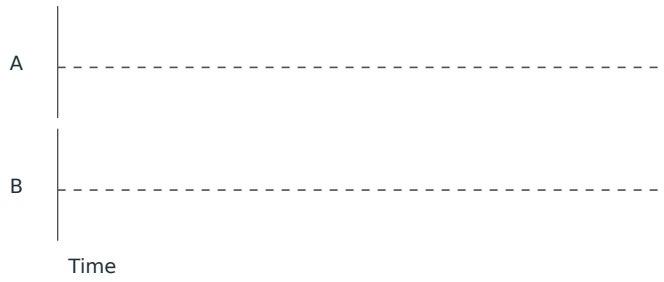
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $T = 1/f$; $v = f\lambda$.
Higher f generally means higher pitch. Amplitude affects intensity under matched conditions, not frequency.

Worked reasoning: A tuning fork makes 440 cycles each second, so its frequency is 440 Hz. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Apply the physics

A tone generator plays 250 Hz and 500 Hz in the same air. Use sound speed 340 m/s to calculate both wavelengths. Separately, double pressure amplitude at fixed frequency. Use $I \propto A^2$ to compare intensity. Label which feature changes in each model.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

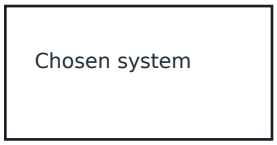
Turning up a speaker’s volume does not normally double a song’s pitch. Explain which wave property mainly changes.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Bodies initially at 340 K and 360 K exchange heat in an isolated system. Explain what happens to total energy and to the temperature-difference opportunity for work at equilibrium.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

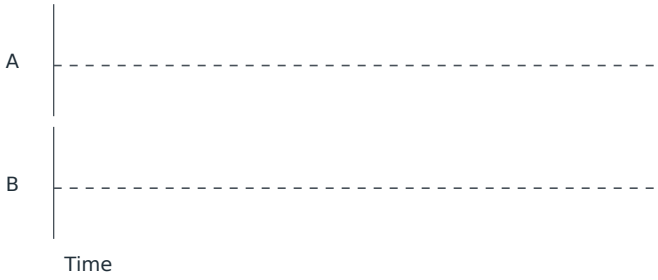


Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

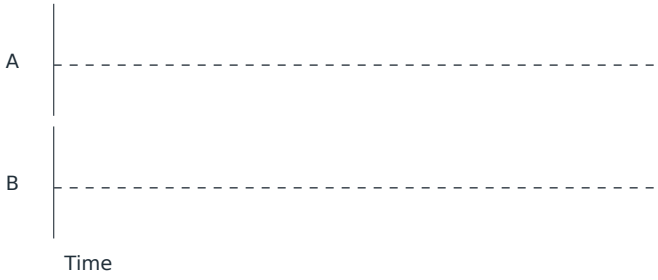
A 500 Hz tone lasts 0.10 s. Find number of cycles and period.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two tones have frequencies 400 and 405 Hz. Using beat frequency as their difference, find the beat rate. Explain why beats are not a new 405-minus 400 Hz pitch from either source alone.



Marzano check - circle: 1 2 3